

Van _____ Surname Stud. #. _____

Voorletters _____ Initials

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Handtekening _____ Signature

Datum _____ Date

Question 1

An object is projected vertically upwards and the height above the ground which it reaches in t seconds, is h , which is given by $h = 20t - 5t^2$.

a) Determine the maximum height reached by the object. **Do not** use the equations of motion to solve this!

b) When is this height reached?

'n Voorwerp word vertikaal opwaarts geprojekteer en die hoogte bo die grond, h , wat dit in t sekondes bereik, word gegee deur $h = 20t - 5t^2$.

a) Bereken die maksimum hoogte wat die voorwerp bereik. **Moet nie** die bewegings-vergelykings gebruik om op te los nie!

b) Wanneer bereik die voorwerp hierdie hoogte?

[4]

Question 2

What is the sum of the following four vectors in unit-vector notation?

Wat is die som van die volgende vier vektore in eenheids-vektor notasie?

$$\vec{A} = (2.00 \text{ m})\hat{i} + (3.00 \text{ m})\hat{j}$$

$$\vec{B} : 4.00 \text{ m, at/by } +1.134 \text{ rad}$$

$$\vec{C} = (-4.00 \text{ m})\hat{i} + (-6.00 \text{ m})\hat{j}$$

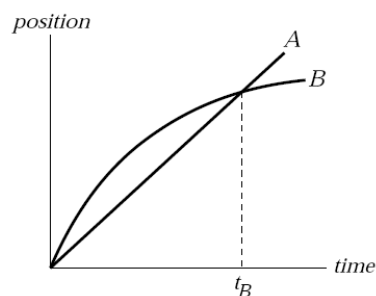
$$\vec{D} : 5.00 \text{ m, at/by } -235^\circ$$

[7]

Question 3

The graph shows position as a function of time for two trains running on parallel tracks. Which of the following is true?

1. At time t_B , both trains have the same velocity.
2. Both trains speed up all the time.
3. Both trains have the same velocity at some time before t_B .
4. Somewhere on the graph, both trains have the same acceleration.



Die grafiek dui die posisie as 'n funksie van tyd aan vir twee treine wat parallel loop. Watter van die volgende is waar?

1. By tydstip t_B , het altwee treine dieselfde snelheid.
2. Altwee treine se spoed neem toe die hele tyd.
3. Altwee treine het dieselfde snelheid op 'n sekere tyd voor t_B .
4. Iewers op die grafiek, het altwee treine dieselfde versnelling.

[3]

Question 4

The position of a particle moving along the x -axis is given by $x = 12t^2 - 2t^3$, where x is in meters and t is in seconds.

- a) Determine the velocity at $t = 3.0$ s.
- b) What is the maximum position reached by the particle along the x -axis?
- c) At what time is this position reached?
- d) What is the particle's acceleration at the time in (c)?

Die posisie van 'n deeltjie wat langs die x -as beweeg word deur $x = 12t^2 - 2t^3$ gegee, waar x in meter is en t in sekondes.

- a) Bepaal die snelheid by $t = 3.0$ s.
- b) Wat is die deeltjie se maksimum posisie langs die x -as?
- c) Hoe lank vat dit om hierdie posisie te bereik?
- d) Wat is die deeltjie se versnelling op die oomblik in (c)?

[8]

THE END / DIE EINDE

Question 3

A person standing at the edge of a cliff throws one ball straight up and another ball straight down at the same initial speed. Which of the two balls will hit the ground below the cliff with the greater speed? Neglect air resistance. Give a short reason for your answer.

1. The one initially thrown upward.
2. The one initially thrown downward.
3. Neither—they both hit at the same speed.

'n Persoon staan op die rant van 'n afgrond en gooi een bal opwaarts en 'n ander bal reguit afwaarts teen dieselfde aanvanklike spoed. Watter een van die twee balle sal die grond tref met die groter spoed? Ignoreer lugweerstand. Gee 'n kort rede vir jou antwoord.

1. *Die een wat aanvanklik opwaarts gegooi is.*
2. *Die een wat aanvanklik afwaarts gegooi is.*
3. *Nie een van die twee nie – albei tref die grond met dieselfde spoed.*

[3]

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Formulas

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$v_{avg} = \frac{1}{2}(v_0 + v)$$

$$v = v_0 + at$$

$$x = x_0 + v_0 t + \frac{1}{2}at^2$$

$$x = x_0 + v_{avg} t$$

$$v^2 = v_0^2 + 2a(x - x_0)$$

$$x - x_0 = \frac{1}{2}(v_0 + v)t$$

$$x - x_0 = vt - \frac{1}{2}at^2$$

$$a = \frac{v^2}{r}$$

$$g = 9,8 \text{ m/s}^2$$

Formules

$$h = \frac{v_0^2 \sin^2 \theta}{2g}$$

$$R = \frac{v_0^2}{g} \sin 2\theta_0$$

$$T = \frac{2v_0 \sin \theta}{g}$$

In the figure, a rescue plane flies at 198km/h and constant height $h = 500$ m toward a point directly over a victim, where a rescue dinghy (inflatable boat) is to land.

The diagram illustrates the geometry of a missile intercept. A missile is launched from a height h with an initial velocity $\vec{v}_0$. The trajectory is a curved path. A line of sight connects the missile to a target on the ground. The angle of elevation is ϕ , and the angle of the line of sight is θ .

- Wat moet vlieënier se gesigslyn (line of sight) se hoek ϕ wees, wanneer die bootjie losgelaat word? [5]

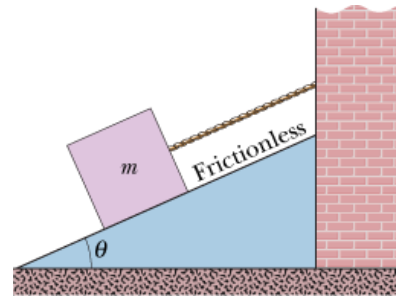
[illegible]

- Wat is die snelheid, $\vec{v}$, van die bootjie in eenheids-vektor notasie (d.w.s. in terme van $\hat{i}, \hat{j}$) soos dit die water tref? [6]

Question 2

In the figure, let the mass of the block be 8,5 kg and the angle θ be 30° . Find the following:

In die figuur het die blok 'n massa van 8,5 kg en die hoek θ is 30° . Bereken die volgende:



- (a) the tension in the cord.
die spankrag in die koord.

[3+2]

- (b) the normal force acting on the block.
die normaalkrag wat op die blok inwerk.

[3]

- (c) If the cord is cut, find the magnitude and direction of the acceleration of the block.
Indien die koord gesny word, bepaal die grootte en rigting van die versnelling van die blok.

[4]

Question 3

In the figure (a), a constant horizontal force $\vec{F}_a$ is applied to block A, which pushes against block B with a 20.0 N force directed horizontally to the right. In Figure (b), the same force $\vec{F}_a$ is applied to block B; now block A pushes on block B with a 10.0 N force directed horizontally to the left. The blocks have a combined mass of 12.0 kg.

In figuur (a), word 'n konstante horisontale krag $\vec{F}_a$ toegepas op blok A, wat dan op blok B druk met 'n krag van 20 N

horisontaal na regs. In figuur (b)

word dieselfde krag $\vec{F}_a$ toegepas op

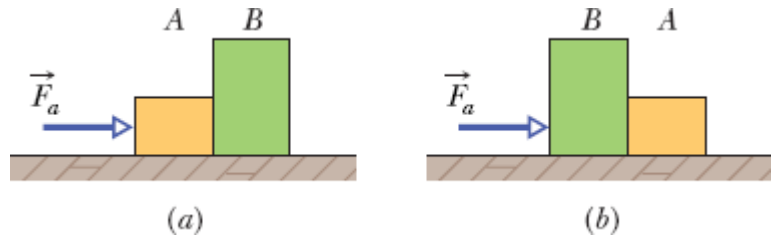
blok B, en nou druk blok A teen

blok B met 'n horisontale krag van

10 N horisontaal na links. Die

blokke het 'n totale massa van 12.0

kg



(a) Calculate the magnitude of their acceleration in Figure (a).

Bereken die grootte van die versnelling in Figuur (a).

[4+2]

(b) Calculate the magnitude of the force $\vec{F}_a$.

Bereken die grootte van die krag $\vec{F}_a$.

[2]

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$$g = 9,8 \text{ m/s}^2$$

$$h = \frac{v_0^2 \sin^2 \theta}{2g}$$

$$R = \frac{v_0^2 \sin 2\theta_0}{g}$$

$$T = \frac{2v_0 \sin \theta}{g}$$

$$T = \frac{2v_0 \sin \theta}{g}$$

$$f = \mu N$$

$$D = \frac{1}{2}C \rho A v^2$$

$$W = \vec{F} \cdot \vec{d} = Fd \cos \phi$$

$$F = -kx$$

$$W = -\frac{1}{2}kx^2$$

Formules

$$P_{avg} = \frac{W}{\Delta t}$$

$$P = \frac{dW}{dt}$$

$$P = \vec{F} \cdot \vec{v}$$

$$U = \frac{1}{2}kx^2$$

$$x_{com} = \frac{1}{M} \sum_{i=1}^n m_i x_i$$

$$\vec{r}_{com} = \frac{1}{M} \sum_{i=1}^n m_i \vec{r}_i$$

$$\vec{p} = m\vec{v}$$

$$\vec{F}_{net} = \frac{d\vec{p}}{dt}$$

Question 1/Vraag 1

A bus rounds a flat corner of radius 9.1 m, at 16 km/h.

'n Bus beweeg om 'n gelyke draai met radius 9.1 m teen 16 km/h.

- a) What is the centripetal acceleration experienced by a short piece of rope hanging from the roof in the bus?

Wat is die sentripetale versnelling wat 'n kort tou ondervind, wat van die dak van die bushang?

[4]

b) What angle with the vertical will be made by the rope?

Wat is die hoek wat die tou maak ten opsigte van 'n vertikale verwysinglyn?

[4+2]

Question 2/Vraag 2

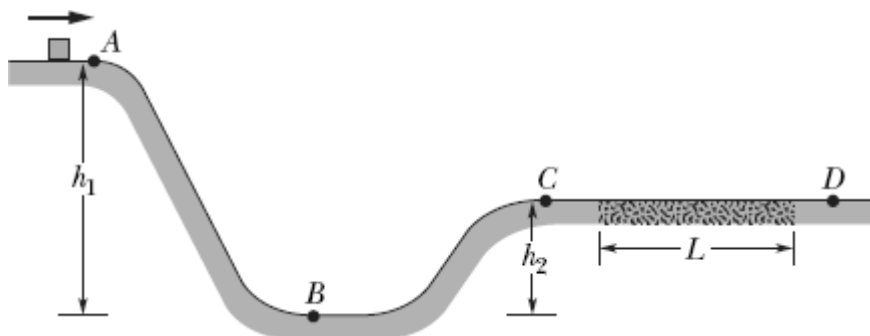
A single force acts on a 3.0 kg particle-like object whose position is given by $x = 3.0t - 4.0t^2 + 1.0t^3$, with x in meters and t in seconds. Find the work done on the object by the force from $t = 0$ to $t = 4.0$ s.

'n Enkel krag werk in op 'n 3.0 kg deeltjie-agtige voorwerp, wat se posisie deur $x = 3.0t - 4.0t^2 + 1.0t^3$ gegee word, waar x in meter is en t in sekondes is. Bepaal die arbeid verrig deur hierdie krag tussen $t = 0$ tot $t = 4.0$ s.

[5]

Question 3/Vraag 3

In the figure, a small block is sent through point A with a speed of 7.0 m/s . Its path is without friction until it reaches the section of length $L = 12 \text{ m}$, where the coefficient of kinetic friction is 0.70 . The indicated heights are $h_1 = 6.0 \text{ m}$ and $h_2 = 2.0 \text{ m}$.



In die figuur word 'n klein

blokkie deur punt A gestuur teen 'n spoed van 7.0 m/s . Die pad wat gevolg word is wrywingloos tot die lengte $L = 12 \text{ m}$, waar die koëffisiënt van kinetiese wrywing 0.70 is. Die hoogtes wat aangedui word is $h_1 = 6.0 \text{ m}$ en $h_2 = 2.0 \text{ m}$.

- a) What is the speed of the block at point B?
Wat is die spoed van die blokkie by punt B?

[3]

- b) What is the speed of the block at point C?
Wat is die spoed van die blokkie by punt C?

[3]

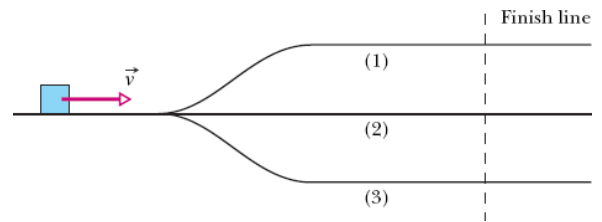
- c) Does the block reach point D? If so, what is its speed at point D; if not, how far through the section with friction does it travel?
Bereik die blokkie punt D? Indien wel, wat is die blokkie se spoed by punt D? Indien nie, hoe ver langs die wrywingsoppervlak beweeg die blokkie?

[5]

Question 4/Vraag 4

In the figure, a horizontally moving block can take three frictionless routes, differing only in elevation, to reach the dashed finish line.

In die figuur, beweeg 'n blok horisontaal en kan drie moontlike roetes neem om by die eindpunt te kom. Die roetes verskil slegs in hoogte.



- a) Rank the routes according to the speed of the block at the finish line – fastest first.
Rangskik die roetes ten opsigte van hul spoed by die eindpunt – vinnigste eerste.

[2]

- b) Rank the routes according to the time taken to reach the finish line – least time first.
Rangskik die roetes ten opsigte van die tyd wat dit sal neem om by die eindpunt te kom – minste tyd eerste.

[2]

Have a nice day / Geniet die day

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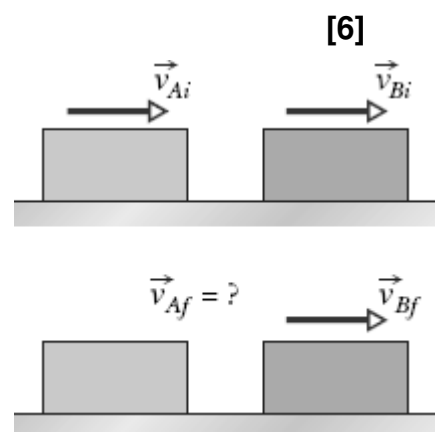
Date _____ Datum

Question 1/Vraag 1

In the Figure, block A (mass 1.6 kg) slides into block B (mass 2.4 kg), along a frictionless surface. The directions of three velocities before (i) and after (f) the collision are indicated; the corresponding speeds are , $v_{Ai} = 5.5$ m/s, $v_{Bi} = 2.5$ m/s and $v_{Bf} = 4.9$ m/s.

Die figuur dui blok A (massa 1.6 kg) wat oor 'n
wrywinglose oppervlak gly in blok B (massa 2.4 kg) vas.
Die rigtings van drie snelhede voor (i) en na (f) die
botsing word aangedui; die ooreenstemmende waardes
van die drie is $v_{Ai} = 5.5$ m/s, $v_{Bi} = 2.5$ m/s and $v_{Bf} = 4.9$
m/s.

- a) What are the velocity v_{Af} ?
Wat is die snelheid v_{Af} ?



[3]

- b) Is the collision elastic? Give reasons for your answer.
Is hierdie 'n elastiese botsing? Gee redes vir jou antwoord

[3]

Question 2/Vraag 2**[5]**

A wheel has a constant angular acceleration of 3.0 rad/s^2 . During a certain 4.0 s interval, it turns through an angle of 120 rad.

'n Wiel het 'n konstante hoekversnelling van 3.0 rad/s^2 . Gedurende 'n sekere 4.0 s interval, draai dit deur 'n hoek van 120 rad.

- a) Show clearly that the angular speed of the wheel at the beginning of the 4.0 s interval is 24 rad/s?

Dui duidelik aan dat die hoeksnelheid van die wiel aan die begin van die 4.0 s interval 24 rad/s was?

[3]

- b) What is the linear speed of a point at the rim of the wheel at the beginning of the 4.0 s interval if the radius is 5 cm?

Wat is die liniêre spoed van 'n punt op die rant van die wiel aan die begin van die 4.0 s interval indien die radius 5 cm is?

[2]

Question 3/Vraag 3**[8]**

The figure shows a uniform disk that can rotate around its center like a merry-go-round. The disk has a radius of 2.00 cm and a mass of 40.0 grams and is initially at rest.

Starting at time $t = 0$, two forces are to be applied tangentially to the rim as indicated, so that at time $t = 1.25$ s the disk has an angular velocity of 250 rad/s

counterclockwise. Force $\vec{F}_1$ has a magnitude of 0.200 N.



Die figuur dui 'n univorme skuif aan wat om die middelpunt kan roteer (soos 'n mallemeule). Die skuif het 'n radius van 2.00 cm en 'n massa van 40.0 g en is aanvanklik met rus. Beginende by $t = 0$, werk twee kragte tangensiaal in op die rant van die skuif soos aangedui, sodat die skuif 'n hoeksnelheid van 250 rad/s antikloksgewys het teen tyd $t = 1.25$ s. Krag $\vec{F}_1$ het 'n grootte van 0.200 N.

a) What is magnitude of F_2 ?

Wat is die grootte van F_2 ?

[5]

b) Determine the angular momentum of the disk at time 1.25 s.

Bepaal die hoekmomentum van die disk teen tyd 1.25 s

[3]

Question 4/Vraag 4**[8]**

A 0.10 kg block oscillates back and forth along a straight line on a frictionless horizontal surface. Its displacement from the origin is given by $x = (10 \text{ cm}) \cos \left[(10 \text{ rad/s})t + \frac{\pi}{2} \text{ rad} \right]$.

'n 0.10 kg blok ossilleer heen en weer in 'n reguitlyn op 'n wrywinglose horisontale oppervlak. Sy verplasing van die oorsprong word gegee deur

$$x = (10 \text{ cm}) \cos \left[(10 \text{ rad/s})t + \frac{\pi}{2} \text{ rad} \right] .$$

- a) Determine the frequency of the oscillations.

Bepaal die frekwensie van die ossillasies.

[2]

- b) Determine the maximum speed acquired by the block.

Bepaal die maksimum spoed wat die blok bereik.

[3]

- c) What force, applied to the block by the spring, results in the given oscillation?

Watter krag word deur die veer op die blok uitgeoefen om hierdie gegewe ossillasie te veroorsaak?

[3]

Enjoy the rest of the day / Geniet die res van die dag

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Question 1

An object is projected vertically upwards and the height above the ground which it reaches in t seconds, is h , which is given by $h = 20t - 5t^2$.

- Determine the maximum height reached by the object. **Do not** use the equations of motion to solve this!
- When is this height reached?

'n Voorwerp word vertikaal opwaarts geprojekteer en die hoogte bo die grond, h , wat dit in t sekondes bereik, word gegee deur $h = 20t - 5t^2$.

- Bereken die maksimum hoogte wat die voorwerp bereik. **Moet nie** die bewegings-vergelykings gebruik om op te los nie!
- Wanneer bereik die voorwerp hierdie hoogte?

[4]

a, b) To determine the possible max height, $dh/dt = 0$.

$$h = 20t - 5t^2$$

$$\frac{dh}{dt} = 0 = 20 - 10t$$

This occurs when $t = 2$ s.

The max height is then

$$h = 20t - 5t^2 = 20(2) - 5(2)^2 = 20 \text{ m}$$

Question 2

What is the sum of the following four vectors in unit-vector notation?

Wat is die som van die volgende vier vektore in eenheids-vektor notasie?

$$\vec{A} = (2.00 \text{ m})\hat{i} + (3.00 \text{ m})\hat{j}$$

$$\vec{B}: 4.00 \text{ m, at/by } +1.134 \text{ rad}$$

$$\vec{C} = (-4.00 \text{ m})\hat{i} + (-6.00 \text{ m})\hat{j}$$

$$\vec{D}: 5.00 \text{ m, at/by } -235^\circ$$

[7]

$$B_x = B \cos 1.134 \text{ rad} = 1.69 \text{ m, } B_y = B \sin 1.134 \text{ rad} = 3.62 \text{ m}$$

$$-235^\circ = +125^\circ$$

$$D_x = D \cos 125^\circ = -2.88 \text{ m, } D_y = D \sin 125^\circ = 4.1 \text{ m}$$

$$R_x = 2.00 + 1.69 - 4.00 - 2.88 = -3.19 \text{ m}$$

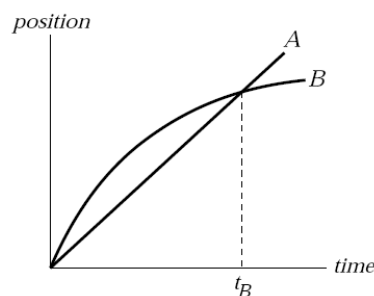
$$R_y = 3.00 + 3.62 - 6.00 + 4.1 = 4.72 \text{ m}$$

$$\vec{R} = (-3.19 \text{ m})\hat{i} + (4.72 \text{ m})\hat{j}$$

Question 3

The graph shows position as a function of time for two trains running on parallel tracks. Which of the following is true?

1. At time t_B , both trains have the same velocity.
2. Both trains speed up all the time.
3. Both trains have the same velocity at some time before t_B .
4. Somewhere on the graph, both trains have the same acceleration.



Die grafiek dui die posisie as 'n funksie van tyd aan vir twee treine wat parallel loop. Watter van die volgende is waar?

1. By tydstip t_B , het altwee treine dieselfde snelheid.
2. Altwee treine se spoed neem toe die hele tyd.
3. Altwee treine het dieselfde snelheid op 'n sekere tyd voor t_B .
4. Iewers op die grafiek, het altwee treine dieselfde versnelling.

[3]

3

Question 4

The position of a particle moving along the x -axis is given by $x = 12t^2 - 2t^3$, where x is in meters and t is in seconds.

- a) Determine the velocity at $t = 3.0$ s.
- b) What is the maximum position reached by the particle along the x -axis?
- c) At what time is this position reached?
- d) What is the particle's acceleration at the time in (c)?

Die posisie van 'n deeltjie wat langs die x -as beweeg word deur $x = 12t^2 - 2t^3$ gegee, waar x in meter is en t in sekondes.

- a) Bepaal die snelheid by $t = 3.0$ s.
- b) Wat is die deeltjie se maksimum posisie langs die x -as?
- c) Hoe lank vat dit om hierdie posisie te bereik?
- d) Wat is die deeltjie se versnelling op die oomblik in (c)?

[8]

- a) $v = \frac{dx}{dt} = 24t - 6t^2$
 $v(3) = 24(3) - 6(3)^2 = 180$ m/s
- b + c) Max/min when $v = 0$ m/s
 $v = 0$ m/s when $t = 0$ s or $t = 4$ s
At $t = 0$ s, $x = 0$ m.
At $t = 4$ s, $x = 64$ m. Max position
- d) $a = \frac{dv}{dt} = 24 - 12t$
 $a(4) = 24 - 12(4) = -24$ m/s²

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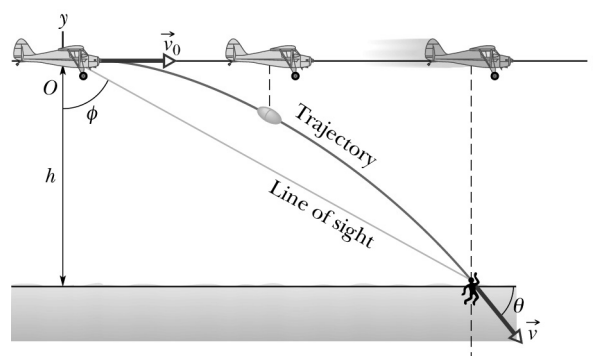
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Question 1

In the figure, a rescue plane flies at 198 km/h and constant height $h = 500$ m toward a point directly over a victim, where a rescue dinghy (inflatable boat) is to land.

In die figuur, 'n reddingsvliegtuig vlieg teen 198 km/h en op 'n konstante hoogte $h = 500$ m in die rigting van 'n slagoffer, waar 'n reddingsbootjie moet land.



- (a) What should be the angle ϕ of the pilot's line of sight to the victim be when the dinghy is released?

Wat moet vlieënier se gesigslyn (line of sight) se hoek ϕ wees, wanneer die bootjie losgelaat word?

[5]

Need to find angle ϕ . From Figure $\phi = \tan^{-1}\left(\frac{x}{h}\right)$, where x is the horizontal distance from the plane to the victim. $h = 500$ m, but x is unknown. Need to determine time taken to fall in order to determine horizontal distance travelled in this time.

Time taken to fall 500 m:

$$y = v_{oy}t + \frac{1}{2}at^2$$

$$-500 \text{ m} = (0)t + \frac{1}{2}(-9.8 \text{ m/s}^2)t^2$$

$$t = \sqrt{\frac{1000}{9.8}} = 10.1 \text{ s}$$

$$v_{ox} = 198 \text{ km/h} = \frac{198000 \text{ m/hr}}{3600 \text{ s/hr}} = 55 \text{ m/s}$$

Horizontal distance covered in this time:

$$x = v_{ox}t + \frac{1}{2}a_x t^2$$

$$= (55.0)t + \frac{1}{2}(0)t^2$$

$$x = 555.5 \text{ m}$$

Determine the angle

$$\phi = \tan^{-1} \frac{555.5}{500} = 48^\circ$$

(b) As the dingy reaches the water, what is its velocity, $\vec{v}$, in unit-vector notation (i.e. in terms of $\hat{i}, \hat{j}$).

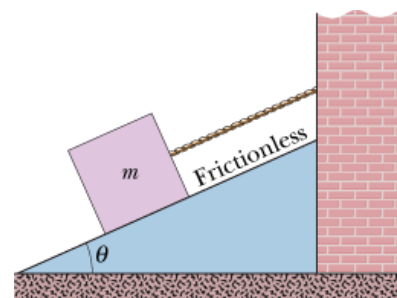
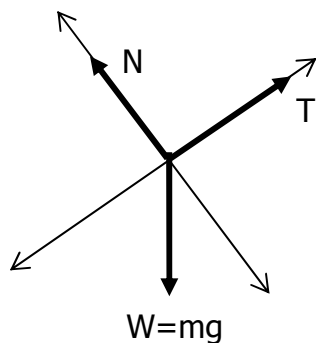
Wat is die snelheid, $\vec{v}$, van die bootjie in eenheids-vektor notasie (d.w.s. in terme van $\hat{i}, \hat{j}$) soos dit die water tref? [6]

$v_x = v_{ox} = 55 \text{ m/s}$
$v_y = v_{oy}t + \frac{1}{2}at^2 = 0(t) + \frac{1}{2}(-9,8)(10,1)^2 = -99,0 \text{ m/s}$
$\vec{v} = (55 \text{ m/s})\hat{i} + (-99 \text{ m/s})\hat{j}$

Question 2

In the figure, let the mass of the block be 8,5 kg and the angle θ be 30° . Find the following:

In die figuur het die blok 'n massa van 8,5 kg en die hoek θ is 30° . Bereken die volgende:



(a) the tension in the cord and
die spankrag in die koord en

[3+2]

Since the acceleration of the block is zero, the components of the Newton's second law equation yield

$$\sum F_y = T - mg \sin \theta = 0$$

Solving the first equation for the tension in the string, we find

$$T = mg \sin \theta = (8,5 \text{ kg})(9,8 \text{ m/s}^2) \sin 30^\circ = 42 \text{ N} .$$

(b) the normal force acting on the block.

die normaalkrag wat op die blok inwerk.

[3]

We solve $F_N - mg \cos \theta = 0$. for the normal force F_N :

$$F_N = mg \cos \theta = (8,5 \text{ kg})(9,8 \text{ m/s}^2) \cos 30^\circ = 72 \text{ N} .$$

- (c) If the cord is cut, find the magnitude and direction of the acceleration of the block.
Indien die koord gesny word, bepaal die grotte en rigting van die versnelling van die blok.

[4]

When the string is cut, $T = 0$ N and the block accelerates.

$$\sum F_y = -mg \sin \theta = -ma$$

The x component of the second law becomes $-mg \sin \theta = ma$, so the acceleration becomes

$$a = -g \sin \theta = -(9.8 \text{ m/s}^2) \sin 30^\circ = -4.9 \text{ m/s}^2.$$

The negative sign indicates the acceleration is down the plane. The magnitude of the acceleration is 4.9 m/s^2 .

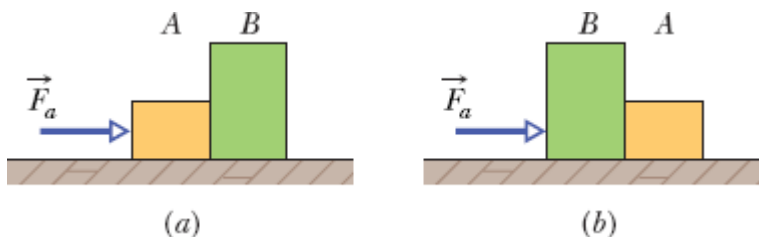
Question 3

In the figure (a), a constant horizontal force $\vec{F}_a$ is applied to block A, which pushes against block B with a 20.0 N force directed horizontally to the right. In Figure (b), the same force $\vec{F}_a$ is applied to block B; now block A pushes on block B with a 10.0 N force directed horizontally to the left. The blocks have a combined mass of 12.0 kg.

In figuur (a), word 'n konstante horisontale krag $\vec{F}_a$ toegepas op blok A, wat dan op blok B druk met 'n krag van 20 N

horisontaal na regs. In figuur (b)

word dieselfde krag $\vec{F}_a$ toegepas op blok B, en nou druk blok A teen blok B met 'n horisontale krag van 10 N horisontaal na links. Die blokke het 'n totale massa van 12.0 kg



- (a) Calculate the magnitude of their acceleration in Figure (a).

Bereken die grootte van die versnelling in Figuur (a).

[4+2]

$\sum F_A = F_a - F_{AB} = m_A a$ $\sum F_B = F_{BA} = m_B a = 20 \text{ N} = F_{AB}$ $F_{BA} = 20 \text{ N}$	$\sum F_B = F_a - F_{BA} = m_B a$ $\sum F_A = F_{AB} = m_A a = F_{BA} = 10 \text{ N}$
For the system total force $F_a = 12a$ $\sum F_A = 12a - 20 = m_A a = 10$ $12a = 30 \text{ N}$ $a = 2.5 \text{ m/s}^{-2}$	

Alternative method:

The (direct) force causing B to have this acceleration in the first figure is twice as big as the (direct) force causing A to have that acceleration. Therefore, B has the twice the mass of A . Since their total is given as 12.0 kg then B has a mass of $m_B = 8.00$ kg and A has mass $m_A = 4.00$ kg. Considering the first figure, $(20.0 \text{ N})/(8.00 \text{ kg}) = 2.50 \text{ m/s}^2$. Of course, the same result comes from considering the second figure $((10.0 \text{ N})/(4.00 \text{ kg}) = 2.50 \text{ m/s}^2)$.

(b) Calculate the magnitude of the force $\vec{F}_a$.

Bereken die grootte van die krag $\vec{F}_a$.

[2]

Using the acceleration determined in part (a)

$$F_a = m_{a+b} a = (12.0 \text{ kg})(2.50 \text{ m/s}^2) = 30.0 \text{ N}$$

THE END / DIE EINDE

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Initials _____ Voorletters

Total/Totaal: _____/30

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she/he is familiar with and accepts
the examination regulations.

Ondergetekende bevestig hiermee dat hy/sy
op hoogte is met die eksamenregulasies
en dat hy/sy hom/haar daaraan onderwerp.

Signature _____ Handtekening Date _____ Datum

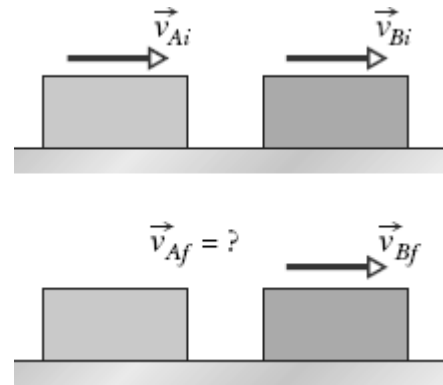
FormulasFormules

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ $v_{avg} = \frac{1}{2}(v_0 + v)$ $v = v_0 + at$ $x = x_0 + v_0 t + \frac{1}{2}at^2$ $x = x_0 + v_{avg} t$ $v^2 = v_0^2 + 2a(x - x_0)$ $x - x_0 = \frac{1}{2}(v_0 + v)t$ $x - x_0 = vt - \frac{1}{2}at^2$ $a = \frac{v^2}{r}$ $g = 9,8 \text{ m/s}^2$	$h = \frac{v_0^2 \sin^2 \theta}{2g}$ $R = \frac{v_0^2}{g} \sin 2\theta_0$ $T = \frac{2v_0 \sin \theta}{g} \quad T = \frac{2v_0 \sin \theta}{g}$ $f = \mu N$ $D = \frac{1}{2}C\rho A v^2$ $W = \vec{F} \cdot \vec{d} = Fd \cos \phi$ $F = -kx$ $W = -\frac{1}{2}kx^2$	$P_{avg} = \frac{W}{\Delta t}$ $P = \frac{dW}{dt}$ $P = \vec{F} \cdot \vec{v}$ $U = \frac{1}{2}kx^2$ $x_{com} = \frac{1}{M} \sum_{i=1}^n m_i x_i$ $\vec{r}_{com} = \frac{1}{M} \sum_{i=1}^n m_i \vec{r}_i$ $\vec{p} = m\vec{v}$ $\vec{F}_{net} = \frac{d\vec{p}}{dt}$
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Question 1/Vraag 1

In the Figure, block A (mass 1.6 kg) slides into block B (mass 2.4 kg), along a frictionless surface. The directions of three velocities before (i) and after (f) the collision are indicated; the corresponding speeds are , $v_{Ai} = 5.5$ m/s, $v_{Bi} = 2.5$ m/s and $v_{Bf} = 4.9$ m/s.

Die figuur dui blok A (massa 1.6 kg) wat in blok B (massa 2.4 kg) gly op 'n wrywinglose oppervlak. Die rigtings van drie snelhede voor (i) en na (f) die botsing word aangedui; die ooreenstemmende drie spoede $v_{Ai} = 5.5$ m/s, $v_{Bi} = 2.5$ m/s and $v_{Bf} = 4.9$ m/s.



- a) What are the velocity v_{Af} ?
Wat is die snelheid v_{Af} ?

[4]

- b) Is the collision elastic?
Wat is die hoek wat die tou maak ten opsigte van 'n vertikale verwysing?

[4]

Question 2/Vraag 2

A wheel has a constant angular acceleration of 3.0 rad/s^2 . During a certain 4.0 s interval, it turns through an angle of 120 rad.

'n Wiel het 'n konstante hoekversnelling van 3.0 rad/s^2 . Gedurende 'n sekere 4.0 s interval, draai dit deur 'n hoek van 120 rad.

- a) What is the angular speed of the wheel at the beginning of the 4.0 s interval?
Wat is die hoeksnelheid van die wiel aan die begin van die 4.0 s interval?

[3]

- b) Assuming that the wheel started from rest, how long has it been in motion at the start of this 4.0 s interval?
Veronderstel dat die wiel oorspronklik met rus was, hoe lang het dit al beweeg voor hierdie 4.0 s interval?

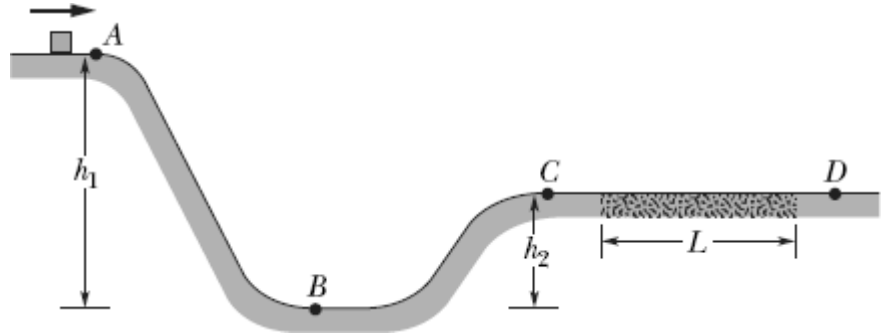
[3]

- c) What is the linear speed of a point at the rim of the wheel at the beginning of the 4.0 s interval?
Wat is die linieêre spoed van 'n punt op die rant van die wiel aan die begin van die 4.0 s interval?

[3]

Question 3/Vraag 3

In the figure, a small block is sent through point A with a speed of 7.0 m/s. Its path is without friction until it reaches the section of length $L = 12$ m, where the coefficient of kinetic friction is 0.70. The indicated heights are $h_1 = 6.0$ m and $h_2 = 2.0$ m.



In die figuur, 'n klein blokkie

word deur punt A gestuur teen 'n spoed van 7.0 m/s. Die pad wat gevolg word is wrywingloos tot die lengte $L = 12$ m, waar die koefisiënt van wrywing 0.70 is. Die hoogtes wat aangedui word is $h_1 = 6.0$ m en $h_2 = 2.0$ m.

- a) What is the speed of the block at point B?
Wat is die spoed van die blokkie by punt B?

[3]

$$\Delta K = \Delta U$$

$$K_B - K_A = U_B - U_A$$

$$\frac{1}{2}mv_B^2 - \frac{1}{2}mv_A^2 = 0 - mgh_1$$

$$v = \sqrt{v_A^2 + 2gh_1}$$

$$= \sqrt{(7.0)^2 + 2(9.8 \text{ m/s}^2)(6.0 \text{ m})}$$

$$= 13 \text{ m/s}$$

If equations of motion were used and the answer was correct then award only **2** marks

- b) What is the speed of the block at point C?
Wat is die spoed van die blokkie by punt C?

[3]

$$\Delta K = \Delta U$$

$$K_C - K_A = U_C - U_A$$

$$\frac{1}{2}mv_C^2 - \frac{1}{2}mv_A^2 = 0 - mgh_1$$

$$v_C = \sqrt{v_A^2 + 2gh_1}$$

$$= \sqrt{(7.0)^2 + 2(9.8 \text{ m/s}^2)(4.0 \text{ m})}$$

$$= 11.3 \text{ m/s}$$

If equations of motion were used and the answer was correct then award only **2** marks

- c) Does the block reach point D? If so, what is its speed there; if not, how far through the section of friction does it travel?

Bereik die blokkie punt D? Indien wel, wat is die blokkie se speed by punt D? Indien nie, hoe ver langs die wryswingsoppervlak beweeg die blokkie?

[5]

$$K_C = \frac{1}{2}mv_C^2 = \frac{1}{2}m(11.3)^2 = 63.85m$$

Work done by friction = ΔK

$$f_k = \mu_k N = \mu_k mg = 0.7(9.8)m$$

Work done by friction across length L :

$$W = -f_k d = -0.7(9.8)(L)m = 82.32m$$

Therefore the block will not reach D .

The distance covered by the block will require ΔK to equal the work done. Therefore:

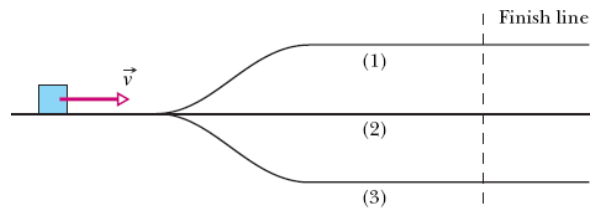
$$\Delta K = W$$

$$63.85m = -f_k d = -0.7(9.8)(L)m$$

$$L = 9.3 \text{ m}$$

Question 4/Vraag 4

In the figure, a horizontally moving block can take three frictionless routes, differing only in elevation, to reach the dashed finish line. Rank the routes according to (a) the speed of the block at the finish line and (b) the travel time of the block to the finish line, greatest first.



In die figuur, beweeg 'n blok horisontaal en kan drie moontlike roetes neem om by die eindpunt te kom. Die roetes verskil slegs in hoogte.

- a) Rank the routes according to the speed of the block at the finish line – fastest first.
Lys die roetes ten opsigte van hul speed by die eindpunt – vinigste eerste.

[2]

3,2,1

- b) Rank the routes according to the time taken to reach the finish line – least time first.
Lys die roetes ten opsigte van die tyd wat dit sal neem om by die eindpunt te kom – minste tyd eerste.

[2]

3,2,1

Have a nice day / Geniet die day

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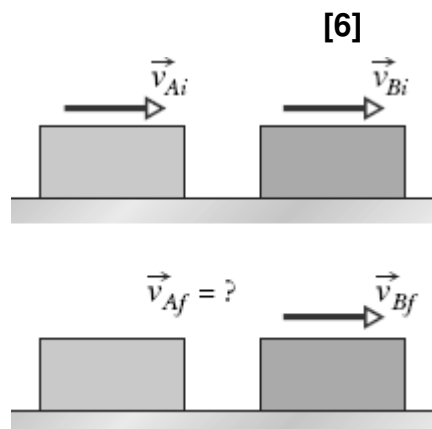
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Signature _____ Handtekening Date _____ Datum

Question 1/Vraag 1

In the Figure, block A (mass 1.6 kg) slides into block B (mass 2.4 kg), along a frictionless surface. The directions of three velocities before (i) and after (f) the collision are indicated; the corresponding speeds are , $v_{Ai} = 5.5$ m/s, $v_{Bi} = 2.5$ m/s and $v_{Bf} = 4.9$ m/s.

Die figuur dui blok A (massa 1.6 kg) wat oor 'n wrywinglose oppervlak gly in blok B (massa 2.4 kg) vas. Die rigtings van drie snelhede voor (i) en na (f) die botsing word aangedui; die ooreenstemmende waardes van die drie is $v_{Ai} = 5.5$ m/s, $v_{Bi} = 2.5$ m/s en $v_{Bf} = 4.9$ m/s.



- c) What are the velocity v_{Af} ?
Wat is die snelheid v_{Af} ?

[3]

$$m_A v_{Ai} + m_B v_{Bi} = m_A v_{Af} + m_B v_{Bf}$$

and

$$\begin{aligned} v_{Af} &= \frac{m_A v_{Ai} + m_B v_{Bi} - m_B v_{Bf}}{m_A} \\ &= \frac{(1.6 \text{ kg})(5.5 \text{ m/s}) + (2.4 \text{ kg})(2.5 \text{ m/s}) - (2.4 \text{ kg})(4.9 \text{ m/s})}{1.6 \text{ kg}} \\ &= 1.9 \text{ m/s.} \end{aligned}$$

The block continues going to the right after the collision.

- d) Is the collision elastic? Give reasons for your answer.
Is hierdie 'n elastiese botsing? Gee redes vir jou antwoord

[3]

To see if the collision is elastic, we compare the total kinetic energy before the collision with the total kinetic energy after the collision. The total kinetic energy before is

$$\begin{aligned} K_i &= \frac{1}{2} m_A v_{Ai}^2 + \frac{1}{2} m_B v_{Bi}^2 \\ &= \frac{1}{2} (1.6 \text{ kg}) (5.5 \text{ m/s})^2 + \frac{1}{2} (2.4 \text{ kg}) (2.5 \text{ m/s})^2 \\ &= 31.7 \text{ J.} \end{aligned}$$

The total kinetic energy after is

$$\begin{aligned}K_f &= \frac{1}{2}m_A v_{Af}^2 + \frac{1}{2}m_B v_{Bf}^2 \\&= \frac{1}{2}(1.6 \text{ kg})(1.9 \text{ m/s})^2 + \frac{1}{2}(2.4 \text{ kg})(4.9 \text{ m/s})^2 \\&= 31.7 \text{ J}.\end{aligned}$$

Since $K_i = K_f$ the collision is found to be elastic.

Question 2/Vraag 2

[5]

A wheel has a constant angular acceleration of 3.0 rad/s^2 . During a certain 4.0 s interval, it turns through an angle of 120 rad .

'n Wiel het 'n konstante hoekversnelling van 3.0 rad/s^2 . Gedurende 'n sekere 4.0 s interval, draai dit deur 'n hoek van 120 rad .

- d) Show clearly that the angular speed of the wheel at the beginning of the 4.0 s interval is 24 rad/s ?

Dui duidelik aan dat die hoeksnelheid van die wiel aan die begin van die 4.0 s interval 24 rad/s was?

[3]

We take $t = 0$ at the start of the interval and take the sense of rotation as positive. Then at the end of the interval with $\alpha = 3.0 \text{ rad/s}^2$

$t = 4.0 \text{ s}$ then $\theta = 120 \text{ rad}$

$$\theta = \omega_0 t + \frac{1}{2}\alpha t^2$$

Then

$$\omega_0 = \frac{\theta - \frac{1}{2}\alpha t^2}{t} = \frac{120 \text{ rad} - \frac{1}{2}(3.0 \text{ rad/s}^2)(4.0 \text{ s})^2}{4.0 \text{ s}} = 24 \text{ rad/s}.$$

- e) What is the linear speed of a point at the rim of the wheel at the beginning of the 4.0 s interval if the radius is 5 cm ?

Wat is die liniêre spoed van 'n punt op die rant van die wiel aan die begin van die 4.0 s interval indien die radius 5 cm is?

[2]

$$v = \omega r$$

$$= (24)(5 \times 10^{-2}) = 1.2 \text{ m/s}$$

Question 3/Vraag 3

[8]

The figure shows a uniform disk that can rotate around its center like a merry-go-round. The disk has a radius of 2.00 cm and a mass of 40.0 grams and is initially at rest.

Starting at time $t = 0$, two forces are to be applied tangentially to the rim as indicated, so that at time $t = 1.25 \text{ s}$ the disk has an angular velocity of 250 rad/s

counterclockwise. Force $\vec{F}_1$ has a magnitude of 0.200 N .

Die figuur dui 'n univorme skuiw aan wat om die middelpunt kan roteer (soos 'n mallemeule). Die skuiw het 'n radius van 2.00 cm en 'n massa van 40.0 g en is aanvanklik



met rus. Beginnende by $t = 0$, werk twee kragte tangensiaal in op die rant van die skuif soos aangedui, sodat die skuif 'n hoeksnelheid van 250 rad/s antiklosgewys het teen tyd $t = 1.25$ s. Krag $\vec{F}_1$ het 'n grootte van 0.200 N.

- a) What is magnitude of F_2 ?
Wat is die grootte van F_2 ?

[5]

$$\tau_{\text{net}} = RF_2 - RF_1 = I\alpha,$$

$$\omega = \omega_0 + \alpha t$$

$$\alpha = \frac{\omega - \omega_0}{t} = \frac{\omega}{t}$$

For a disk: $I = \frac{1}{2}MR^2$

Solve for F_2 from first equn:

$$\begin{aligned} F_2 &= \frac{1}{2}MR \left(\frac{\omega}{t} \right) + F_1 = \frac{1}{2}(0.04)(0.02) \left(\frac{250}{1.25} \right) + 0.200 \text{ N} \\ &= 0.28 \text{ N} \end{aligned}$$

- b) Determine the angular momentum of the disk at time 1.25 s.
Bepaal die hoekmomentum van die disk teen tyd 1.25 s

[3]

$$\begin{aligned} L &= I\omega = \left(\frac{1}{2}MR^2 \right) 250 \text{ rad/s} \\ &= \left(\frac{1}{2}(40 \times 10^{-3})(2 \times 10^{-2})^2 \right) 250 \text{ rad/s} \\ &= 0.002 \text{ kg} \cdot \text{m}^2 \text{s}^{-1} \end{aligned}$$

Out of the page

Question 4/Vraag 4

[8]

A 0.10 kg block oscillates back and forth along a straight line on a frictionless horizontal surface. Its displacement from the origin is given by $x = (10 \text{ cm}) \cos \left[(10 \text{ rad/s})t + \frac{\pi}{2} \text{ rad} \right]$

'n 0.10 kg blok ossilleer heen en weer in 'n reguitlyn op 'n wrywinglose horisontale oppervlak. Sy verplasing van die oorsprong word gegee deur

$$x = (10 \text{ cm}) \cos \left[(10 \text{ rad/s})t + \frac{\pi}{2} \text{ rad} \right]$$

- c) Determine the frequency of the oscillations.
Bepaal die frekwensie van die ossillasies.

[2]

$$\omega = 10 \text{ rad/s}$$

$$f = \omega / 2\pi = 1.6 \text{ Hz}$$

- d) Determine the maximum speed acquired by the block.
Bepaal die maksimum spoed wat die blok bereik.

[3]

$$v_m = \omega x_m \text{ and } x_m = 10 \text{ cm}$$

then $v_m = (10 \text{ rad/s})(10 \text{ cm}) = 100 \text{ cm/s}$ or 1.0 m/s

- e) What force, applied to the block by the spring, results in the given oscillation?
Watter krag word deur die veer op die blok uitgeoefen om hierdie gegewe ossillasie te veroorsaak?

[3]

$$\omega = \sqrt{\frac{k}{m}} \Rightarrow k = (0.10 \text{ kg})(10 \text{ rad / s})^2 = 10 \text{ N / m.}$$

Thus, Hooke's law gives $F = -kx = -10x$

Enjoy the rest of the day / Geniet die res van die dag